

✓ Задача №5: Кодирование категориального признака с использованием One-Hot Encoding

Для выполнения этой задачи мы выберем категориальный признак `Sex` из набора данных `Titanic` и применим к нему метод `One-Hot Encoding`.

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

print("Original DataFrame head:")
display(df.head())
```

Original DataFrame head:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

✓ One-Hot Encoding признака 'Sex'

Применим `pd.get_dummies` для выполнения One-Hot Encoding для признака `Sex`.

```
# Perform one-hot encoding on the 'Sex' column
df_encoded = pd.get_dummies(df, columns=['Sex'], drop_first=True)

print("DataFrame after One-Hot Encoding for 'Sex' (with 'Sex_male' column):")
display(df_encoded.head())
```

DataFrame after One-Hot Encoding for 'Sex' (with 'Sex_male' column):

	PassengerId	Survived	Pclass	Name	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Sex_male
0	1	0	3	Braund, Mr. Owen Harris	22.0	1	0	A/5 21171	7.2500	NaN	S	True
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1	0	PC 17599	71.2833	C85	C	False
2	3	1	3	Heikkinen, Miss. Laina	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	False
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	0	113803	53.1000	C123	S	False
4	5	0	3	Allen, Mr. William Henry	35.0	0	0	373450	8.0500	NaN	S	True

✓ Задача №25: Обнаружение и удаление выбросов на основе межквартильного размаха (IQR)

Для выполнения этой задачи мы выберем числовой признак `Fare` из набора данных `Titanic` и проведем обнаружение и удаление выбросов на основе межквартильного размаха (IQR).

```
# For outlier detection, we will use the original dataframe or a copy if modifications were made above.
# Let's use the df_encoded for continuity, assuming we want to apply this after encoding.

# Calculate Q1, Q3, and IQR for the 'Fare' column
Q1 = df_encoded['Fare'].quantile(0.25)
Q3 = df_encoded['Fare'].quantile(0.75)
IQR = Q3 - Q1

# Define bounds for outliers
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print(f"Q1 (Fare): {Q1}")
print(f"Q3 (Fare): {Q3}")
print(f"IQR (Fare): {IQR}")
print(f"Lower Bound (Fare): {lower_bound}")
print(f"Upper Bound (Fare): {upper_bound}")

# Identify outliers
outliers = df_encoded[(df_encoded['Fare'] < lower_bound) | (df_encoded['Fare'] > upper_bound)]

print("\nNumber of outliers detected in 'Fare' column:", len(outliers))
print("\nOutliers in 'Fare' column:")
display(outliers[['Fare']].head())
```

Q1 (Fare): 7.9184
Q3 (Fare): 31.0
IQR (Fare): 23.0896
Lower Bound (Fare): -26.724
Upper Bound (Fare): 65.6344

Number of outliers detected in 'Fare' column: 116

Outliers in 'Fare' column:

	Fare
1	71.2833
27	263.0000
44	446.5000

31 140.3200
34 82.1708
52 76.7292

```
# Remove outliers from the DataFrame
df_no_outliers = df_encoded[(df_encoded['Fare'] >= lower_bound) & (df_encoded['Fare'] <= upper_bound)]

print("\nDataFrame head after removing outliers from 'Fare' column:")
display(df_no_outliers.head())
print(f"Original DataFrame shape: {df_encoded.shape}")
print(f"DataFrame shape after outlier removal: {df_no_outliers.shape}")
```

DataFrame head after removing outliers from 'Fare' column:

	PassengerId	Survived	Pclass	Name	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	Sex_male
0	1	0	3	Braund, Mr. Owen Harris	22.0	1	0	A/5 21171	7.2500	NaN	S	True
2	3	1	3	Heikkinen, Miss. Laina	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	False
3	4	1	1	Futelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1	0	113803	53.1000	C123	S	False
4	5	0	3	Allen, Mr. William Henry	35.0	0	0	373450	8.0500	NaN	S	True
5	6	0	3	Moran, Mr. James	NaN	0	0	330877	8.4583	NaN	Q	True

Original DataFrame shape: (891, 12)
DataFrame shape after outlier removal: (775, 12)